

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

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Finding Reasoning Paths :

- Anatomical Snuffbox**: The anatomical snuff box is a palpable depression on the dorsal aspect of the hand , located between the thumb and the lateral edge of the hand . It serves as an anatomical landmark for the radial artery and the extensor pollicis longus muscle . A penetrating injury in this region could damage nearby structures , particularly the radial artery .
- Vascular Anatomy**: The radial artery is a major branch of the brachial artery and runs through the anatomical snuff box . It is located superficially in this region , making it susceptible to injury from penetrating wounds .
- Hemodynamic Stability**: Severe bleeding from a stab wound in the anatomical snuff box is most commonly due to damage to the radial artery , as it is the largest and most superficial artery in this area . Other structures , such as the extensor pollicis longus , are more superficial but are not primary sources of significant bleeding .

Reasoning Process :

- Clinical Presentation**: The patient has severe bleeding from a penetrating stab wound in the anatomical snuff box . This region is anatomically significant because it is surrounded by the radial artery , which is the most common structure to cause significant bleeding in such injuries .
- Anatomical Landmarks**: The anatomical snuff box is divided into three compartments :
 - Radial side**: Contains the radial artery and the extensor pollicis longus muscle .
 - Palmar side**: Contains the superficial and deep palmar arches .
 - Longitudinal side**: Contains the superficial and deep venous plexus .Among these , the radial artery is the most critical and largest artery in this region , making it the most likely source of severe bleeding .

- Vascular Injury in the Anatomical Snuff box** :
 - The radial artery is superficial in the anatomical snuff box and runs along the lateral aspect of the wrist . It is easily accessible and vulnerable to injury from penetrating wounds .
 - Damage to the radial artery can lead to significant hemodynamic instability due to its role as a major collateral artery supplying the hand .

- Exclusion of Other Structures** :
 - The **extensor pollicis longus** is a muscle in the anatomical snuff box but is not a major artery . It is more superficial and less likely to cause severe bleeding compared to the radial artery .
 - Other structures , such as the superficial and deep venous plexus or the palmar arches , are not typically associated with massive bleeding .

- Management** :
 - If the radial artery is damaged , the patient is at risk of severe hemodynamic instability . Immediate intervention , such as surgical repair or occlusion of the wound , is critical to prevent life-threatening blood loss .

Conclusion :

The radial artery is the structure most likely responsible for the excessive bleeding . The anatomical location of the wound and the vascular anatomy of the region strongly support this conclusion .

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< answer >

Radial artery

</answer>

